(A) Functionality Completion Task

Question:

"… addresses the need to differentiate imports that occur only within [MASK1] blocks from standard runtime imports. The implementation adds a new [MASK2] attribute to module nodes to track type-checking imports separately from regular dependencies, utilizes the [MASK3] utility function to detect when imports occur within type checking contexts, and introduces a [MASK4] edge type with corresponding visual representations across all supported output formats… Fill in the [MASK] placeholders with the correct functionality details?"

Answer:

[MASK1]: TYPE_CHECKING [MASK2]: type_depends [MASK3]: in_type_checking_block [MASK4]: TYPE_DEPENDENCY

(B) Related Source Code

pylint/pyreverse/diagrams.py

```
[242] class PackageDiagram(ClassDiagram)
[281]   def add_from_depend(self, node: nodes.ImportFrom, from_module: str):
[289]       if not in_type_checking_block(node):
[290]               package.depends.append(from_module)
[294]   def extract_relationships(self) -> None:
[319]       self.add_relationship(package_obj, dep, "type_depends")
```

inherit *caller/callee*

pylint/checkers/utils.py

```
[1962] def in_type_checking_block(node: nodes.NodeNG) -> bool:
[1980]     if ancestor.test.attrname != "TYPE_CHECKING":
[1981]         continue
```

pylint/pyreverse/diagrams.py

```
[80] class ClassDiagram(Figure, FilterMixIn):
[94]   def get_relationships(self, role: str) -> Iterable[Relationship]:
```

caller/callee

pylint/pyreverse/writer.py

```
[56] def write_packages(self, diagram: PackageDiagram) -> None:
[95]    for rel in diagram.get_relationships("type_depends"):
[99]        self.printer.emit_edge(from_id, to_id, type_=EdgeType.TYPE_DEPENDENCY)
```

: Reference Answer **: Relationship**

(C) Retrieved Software Documentation

Chat:

pylint/pyreverse/diagrams.py

class PackageDiagram(ClassDiagram):

PackageDiagram provides a structured way to model and analyze the architecture of a software system at the package level ... Example: diagram.add_object("core", core_module_node) diagram.add_object("utils", utils_module_node) …

pylint/pyreverse/diadefslib.py

class DefaultDiadefGenerator(LocalsVisitor, DiaDefGenerator):

To provide a visual representation of a project's structure and relationships between its components...

RepoAgent:

pylint/pyreverse/diagrams.py

class PackageDiagram(ClassDiagram):

The PackageDiagram class is utilized in the pyreverse ...The **"in_type_checking_block"** function is crucial for determining whether an import is within **"TYPE_CHECKING"** ...

pylint/pyreverse/diagrams.py

def get_relationships(self, role: str):

This function retrieves and sorts relationships of a specified role from ... EdgeType.**TYPE_DEPENDENCY** for **"type_depends"** relationships ...

(D) Software Documentation Evaluation

LLM-as-a-judge Evaluation Method

Chat:

Completeness Score: 5
Usefulness Score: 5
Clarity Score: 5

RepoAgent:

Completeness Score: 5
Usefulness Score: 5
Clarity Score: 5

SWD-Bench's Evaluation Strategy

Chat:

[MASK1]: type_check ✗
[MASK2]: type_checking_imports ✗
[MASK3]: is_type_checking_import ✗
[MASK4]: TYPE_CHECKING_IMPORT ✗

RepoAgent:

[MASK1]: TYPE_CHECKING ✓
[MASK2]: type_depends ✓
[MASK3]: in_type_checking_block ✓
[MASK4]: TYPE_DEPENDENCY ✓